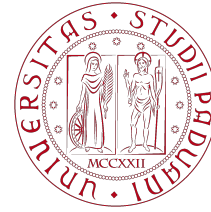


# Final Report

Physics of Complex Networks: Structure and Dynamics



UNIVERSITÀ  
DEGLI STUDI  
DI PADOVA

Areas of physics by complexity



Newton's  
Mechanics

Electro-  
Magnetism

Special  
Relativity

Quantum Mechanics  
General Relativity

Quantum  
Field Theory

Complexity  
Science

## Epidemics on temporal networks & the AS-level Internet from BGP

Venturato, Giancarlo

Last update: July 5, 2026

# Contents

---

|          |                                                    |          |
|----------|----------------------------------------------------|----------|
| <b>1</b> | <b>Epidemic spreading on temporal networks</b>     | <b>1</b> |
| 1.1      | Models and mean-field thresholds . . . . .         | 1        |
| 1.2      | Synthetic temporal topologies and method . . . . . | 1        |
| 1.3      | Results . . . . .                                  | 2        |
| <b>2</b> | <b>Bibliography</b>                                | <b>5</b> |

# 1 | Epidemic spreading on temporal networks

---

**Task leader(s):** *Giancarlo Venturato*

## 1.1 | Models and mean-field thresholds

---

We study the four canonical compartmental models on *time-varying* contact networks. A node is Susceptible (S), Exposed (E, infected but not yet infectious), Infectious (I) or Removed (R). In discrete time, on every contact of snapshot  $t$  an S–I pair transmits with probability  $\beta$ ; an infectious node leaves state I with probability  $\mu$ , and in SEIR an exposed node matures with probability  $\sigma$ . The four models are the same core loop restricted to different states [3]: SI (no recovery), SIS (I→S), SIR (I→R) and SEIR (S→E→I→R).

The order parameter is the endemic prevalence (SI/SIS) or the final epidemic size (SIR/SEIR), which is non-zero only above a critical ratio  $(\beta/\mu)_c$ . On the *activity-driven* temporal network the heterogeneous mean-field threshold is governed by the *activity* moments and not by the aggregated degree [4],

$$\left(\frac{\beta}{\mu}\right)_c^{\text{AD}} = \frac{1}{m(\langle a \rangle + \sqrt{\langle a^2 \rangle})}, \quad (1.1)$$

whereas treating the whole observation window as a single *static* graph gives the classical heterogeneous- (HMF) and quenched- (QMF) mean-field results evaluated on that aggregate,

$$\left(\frac{\beta}{\mu}\right)_c^{\text{HMF}} = \frac{\langle k \rangle}{\langle k^2 \rangle}, \quad \left(\frac{\beta}{\mu}\right)_c^{\text{QMF}} = \frac{1}{\lambda_{\max}(A)}. \quad (1.2)$$

Because the aggregate accumulates every contact ever made,  $\langle k^2 \rangle$  and  $\lambda_{\max}$  are large and the static threshold is orders of magnitude below Eq. (1.1): assuming that contacts are simultaneous systematically over-estimates spreading. Making this contrast explicit is the aim of the task.<sup>1</sup>

## 1.2 | Synthetic temporal topologies and method

---

A temporal network is a sequence of  $T$  single-step snapshots. We use two structurally different synthetic generators, so the models are exercised across a variety of topologies.

---

<sup>1</sup>The simulator uses discrete-time per-step probabilities  $(\beta, \mu)$ , which map onto the continuous rates of Eqs. (1.1)–(1.2) for  $\beta, \mu \ll 1$ ; the agreement in Fig. 1.1(b) confirms the mapping in the regime studied.

(i) The *activity-driven* (AD) model [4]: each node draws an activity  $a_i \propto x^{-\gamma}$ ,  $x \in [\varepsilon, 1]$ ; at every step it is active with probability  $a_i$  and, if so, fires  $m$  links to random partners. There is no fixed backbone. (ii) An *edge-activated* process on a fixed *scale-free* (Barabási–Albert) or *homogeneous* (Erdős–Rényi) backbone whose edges switch on independently with probability  $p$  each step, so the aggregate is the backbone itself. As a real-world test we additionally use the empirical *SocioPatterns Hypertext 2009* face-to-face contact network [2] (113 attendees, 20 s resolution,  $\sim 59$  h, binned into 5-min snapshots). Against each temporal run we compare two null models built from the same data: the *aggregated static* network (all edges present at every step) and a *time-shuffled* surrogate (contacts reassigned to random steps, destroying temporal order while preserving the aggregate and the total contact volume). A single seeded stochastic engine (Python + igraph) runs all four models on any of these sequences; results are averaged over 15–40 realisations.<sup>2</sup> Synthetic default:  $N = 1500$ ,  $T = 800$ ,  $m = 1$ ,  $\gamma = 2.1$ ,  $\mu = 0.02$ .

### 1.3 | Results

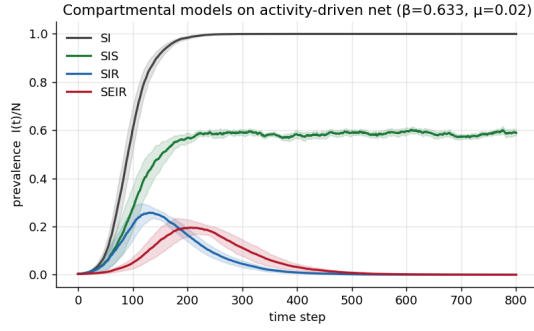
Figure 1.1(a) shows the four models on one AD network above threshold: SI saturates, SIS relaxes to an endemic plateau, and SIR/SEIR are single outbreaks, with SEIR delayed by the latent stage. Panel (b) locates the SIS threshold: the simulated prevalence departs from zero exactly at the activity-driven prediction  $\beta_c^{\text{AD}}$  of Eq. (1.1) (Table 1.1), far above the static-network value  $\beta_c^{\text{HMF}}$ . The time-shuffled curve coincides with the temporal one, confirming that in a memoryless activity-driven model the *ordering* of contacts is immaterial – the relevant effect is concurrency, not order. Panel (c) isolates that concurrency effect for SIR: collapsing the timeline into the aggregated static graph turns a slow, bounded outbreak (peak prevalence  $\approx 0.18$ ) into an explosive one (peak  $\approx 0.98$ , final size 1). Panel (d) exercises the models across backbone *structure*: with the same mean degree  $\langle k \rangle = 4$ , the scale-free BA backbone (hubs) yields a markedly larger outbreak than the homogeneous ER one (peak 0.12 vs. 0.06), while both remain far below their common aggregated-static limit. Treating a temporal network as static is therefore not a harmless approximation: it moves the epidemic threshold by nearly three orders of magnitude and grossly inflates the predicted outbreak.

Finally, Fig. 1.2 applies the same analysis to the empirical Hypertext 2009 contact network. The picture is qualitatively identical – aggregation inflates a bounded outbreak (final size 0.20) into a near-complete one (0.999) – but with one instructive difference from the synthetic case: *time-shuffling now lowers the outbreak* (final size  $0.11 < 0.20$ ), because real contacts carry temporal correlations (bursts, group co-presence) that a memoryless activity-driven model lacks. Both the timing *and* the ordering of contacts matter in real data.

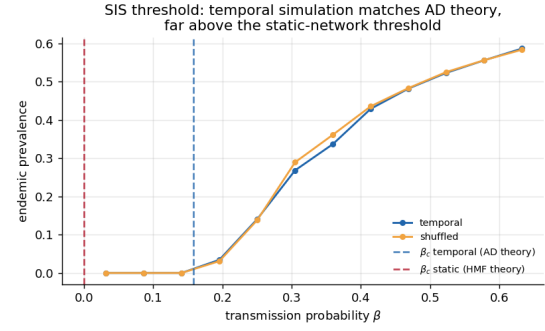
<sup>2</sup>Generators, engine, thresholds and output edge lists are in `code/task_22/` and `data/task_22/`; see Holme & Saramäki [1] for the temporal-network framework.

|                              | $\beta_c^{\text{AD}}$ (temporal) | $\beta_c^{\text{HMF}}$ (static) | $\beta_c^{\text{QMF}}$ (static) |
|------------------------------|----------------------------------|---------------------------------|---------------------------------|
| $(\beta/\mu)_c$              | 7.91                             | $9.5 \times 10^{-3}$            | $1.1 \times 10^{-2}$            |
| $\beta_c$ (at $\mu = 0.02$ ) | 0.158                            | $1.9 \times 10^{-4}$            | $2.2 \times 10^{-4}$            |

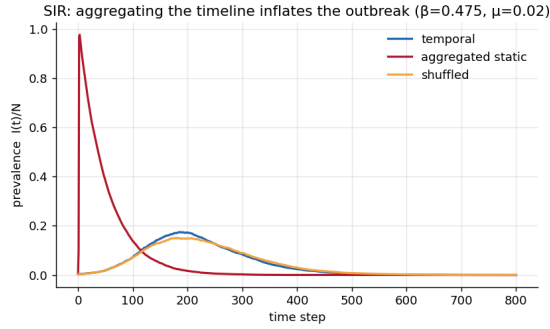
Table 1.1: Critical points for the default AD network ( $\langle a \rangle = 0.041$ ,  $\langle a^2 \rangle = 7.4 \times 10^{-3}$ ; aggregate  $\langle k \rangle = 61$ ,  $\langle k^2 \rangle = 6.4 \times 10^3$ ). The temporal threshold is  $\sim 800\times$  the static one.



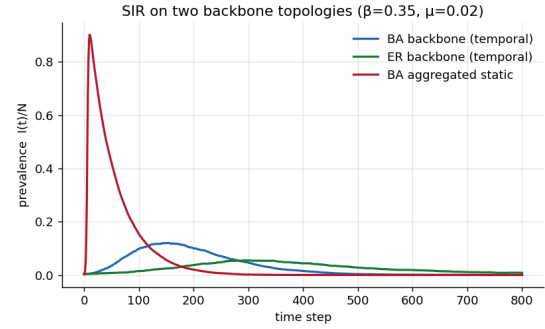
(a) Four models on the AD network



(b) SIS threshold vs. theory



(c) SIR: temporal vs. aggregated (AD)



(d) SIR: BA vs. ER backbones

Figure 1.1: Epidemics on temporal networks. (a) SI/SIS/SIR/SEIR prevalence on one activity-driven network ( $\beta = 4\beta_c^{\text{AD}}$ ); bands are  $\pm 1\sigma$  over realisations. (b) SIS endemic prevalence vs.  $\beta$ : the temporal simulation matches  $\beta_c^{\text{AD}}$  (dashed) and lies far above  $\beta_c^{\text{HMF}}$ ; time-shuffled  $\approx$  temporal. (c) Aggregating the timeline (red) inflates the SIR outbreak relative to the temporal dynamics (blue). (d) Backbone structure at fixed  $\langle k \rangle$ : the scale-free BA network spreads more than the homogeneous ER one.

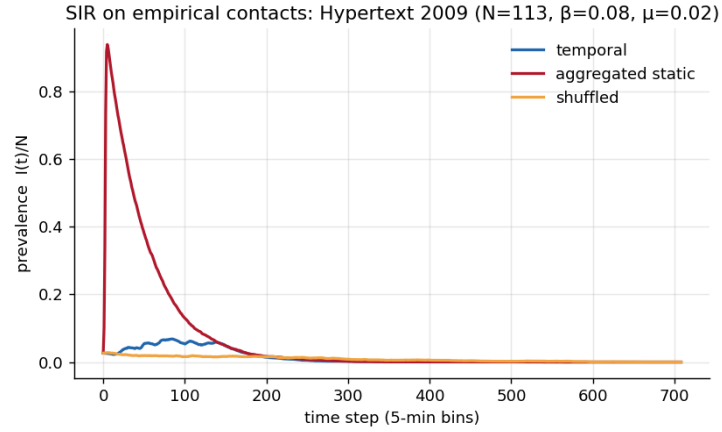


Figure 1.2: Application to empirical data: SIR on the SocioPatterns Hypertext 2009 face-to-face contact network. Aggregating the timeline (red) inflates the outbreak from a final size of 0.20 (temporal) to 0.999; unlike the memoryless synthetic models, time-shuffling (orange) *reduces* spreading (0.11), revealing temporal correlations in the real contacts.

## 2 | Bibliography

---

- [1] Petter Holme and Jari Saramäki. Temporal networks. *Physics Reports*, 519(3): 97–125, 2012.
- [2] Lorenzo Isella, Juliette Stehlé, Alain Barrat, Ciro Cattuto, Jean-François Pinton, and Wouter Van den Broeck. What’s in a crowd? analysis of face-to-face behavioral networks. *Journal of Theoretical Biology*, 271(1):166–180, 2011.
- [3] Romualdo Pastor-Satorras, Claudio Castellano, Piet Van Mieghem, and Alessandro Vespignani. Epidemic processes in complex networks. *Reviews of Modern Physics*, 87(3):925–979, 2015.
- [4] Nicola Perra, Bruno Gonçalves, Romualdo Pastor-Satorras, and Alessandro Vespignani. Activity driven modeling of time varying networks. *Scientific Reports*, 2:469, 2012.